

Inside stars: Modelling convective boundary mixing

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Abstract

This report describes the background, methodology, and aims of the project. Within stars convective motions can penetrate into adjacent radiative zones, and the modelling of these motions remains an open question. The project aims to analyse the overshooting in both the context of dimensionality of simulation used, and the effect of rotation. To do this the project will apply extreme value theory to solar simulations calculated using Multidimensional Stellar Implicit Code (MUSIC).

1 Introduction

Stars are a vital component within the night sky, and are one of the major sources of information about the Universe. From cepheid variables, which give insight into the cosmic distances within the universe, to main-sequence stars, which help build our understanding of our own solar system. There are many different types of stars, and to help our understanding of them involves increasing the accuracy of current models of the internal structure.

The interiors of stars can be split into convective and radiative zones, and the interface between them is not a solid barrier due to convective motions crossing the boundary. The extent of this mixing can have profound effects on the star and our observations, and is more specifically known as overshooting or penetration. From Lithium abundances in the upper atmosphere, estimation in ages of stars (Barnes 2003), temperatures, and luminosity, all of which affect our understanding of stellar evolution. Constraining this process is therefore vital for producing accurate stellar models.

However due to the complex, non-linear nature of motions within stars, analytic solutions are not possible and the use of computational simulations within the field is extensive. Using the MUSIC code (Viallet et al. 2011,

2016; Goffrey et al. 2017), two- and three-dimensional solar models have been produced (Vlaykov et al. 2022). The primary goal of this project is to use these simulations to quantify the extent of convective boundary mixing. A key aspect of this will be to compare the differences in the two- and three-dimensional models. This project will also examine the effect of stellar rotation rates on overshooting, an area where theoretical predictions remain uncertain. To achieve this a methodology pioneered by Pratt et al. (2017) will be utilised in the use of extreme value statistics to describe penetration from rare, powerful plumes. The results from this analysis can then be used to inform and calibrate the parametrisation of convective boundary mixing in one-dimensional (1D) stellar evolution codes, contributing to the "321D" link. This will allow the predictions of our simulations to be tested against observational data, both solar and extrasolar.

The purpose of this report is to discuss the physical background, previous work in the area and present the methodology to be used in the future of the project.

2 Background

2.1 Schwarzschild & Ledoux criteria

Given the differentiation of the interior of stars into a convective, and radiative regime, Karl Schwarzschild developed the Schwarzschild criteria as a method of determining the structure of regions in stellar interiors (Schwarzschild 1906). The criteria assumes a homogeneous composition, and is generally given in modern astrophysical contexts in the form of

$$\nabla_{\text{rad}} < \nabla_{\text{ad}} \quad (1)$$

(Kippenhahn et al. 2012), where the left-hand side is the radiative temperature gradient, defined as the temperature change, as a function of pressure, in equilibrium with radiation and the right-hand side therefore denotes the temperature change a fluid blob will experience when moved with no heat exchange. The inequality of the criteria, means it is valid within a radiative region of a star, but invalid in a convective region.

If the criteria predicts a stable region, energy is transported solely by radiation, $\nabla = \nabla_{\text{rad}}$. However if instability is predicted, then the general assumption is $\nabla \approx \nabla_{\text{ad}}$, which holds apart from in the case of overradiabacity.

A more generalised form of the Schwarzschild criterion, which relaxes the assumption of homogeneity, can be used in regions where the composition changes rapidly. This form is called the Ledoux criterion,

$$\nabla_{\text{rad}} < \nabla_{\text{ad}} + \frac{\varphi}{\delta} \nabla_{\mu}, \quad (2)$$

where ∇_{μ} is the chemical gradient, φ is the density change with composition, and δ is the density change with temperature (Ledoux 1947).

Both the criteria above assume adiabatic energy transfer.

The point at which the criteria reach equality defines the convective boundary

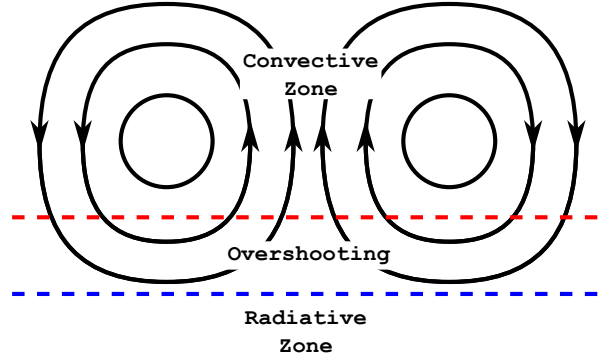


Figure 1: Shows the convective cells which penetrate past the Schwarzschild criterion (red), and the extent of overshooting into the radiative zone (blue).

2.2 Overshooting & Penetration

Within a star convective motions can cross the boundary, classically defined by the Schwarzschild criterion, into the neighbouring radiative region. Figure 1 shows how motions can penetrate into the stratified radiative zone. In an astrophysical context this is commonly interchangeably named both penetration and overshooting (Zahn 1991). Penetration can be found in many other areas of nature (Deardorff et al. 1969), and has been studied for much longer in other areas of research than has been in stars.

Given the importance of penetration on the life of a star, it is vital to come up with a theoretical understanding of the underlying physics involved in the process. However it is not an easy characterisation, as it has been shown mixing length theory alone cannot describe penetrative plumes (Renzini 1987).

Working from the techniques used to analyses overshooting in convective rolls (van Ballegooijen 1982), Zahn's theory then expands to consider the effect of heat flux for a ballistic plumes approach (Zahn 1991). Zahn assumes the penetrative region is characterised by efficient convective energy transfer, meaning $\nabla = \nabla_{\text{ad}}$, and that the root-mean-square (rms) velocity of the motions, $W = 0$, at the extent of this region. The justification for these assumptions come from the thinness of the thermal boundary layer, found between the

extent of total adiabaticity and total radiative transfer, and the low velocities within this region. From these assumptions, Zahn derives a scaling law for the penetration for both convective envelopes and convective cores. The more relevant scheme, for us, of a scaling law is for convective envelopes, where the penetration length,

$$\frac{L_p}{H_p} \propto W_0^{3/2} f^{1/2} \quad (3)$$

where L_p/H_p is the penetration length in units of the pressure scale height, H_p . W_0 is the initial rms velocity at the convective boundary, defined by the Schwarzschild criteria (1). Finally f represents the filling factor, the fraction of the area occupied by down flows.

The inclusion of the filling factor, has been suggested to be the dominant scaling factor over the exit velocity (Brummell et al. 2002; Pratt et al. 2020). However the measurement of this filling factor is not clear (Zahn 1991), and multiple differing definitions have been suggested, as discussed by (Dethero et al. 2024). Due to this other parameters have also been suggested that compare flows at multiple radii rather than at a singular point (Anders et al. 2022).

2.3 Observational evidence

There are several observational methods that provide evidence for convective boundary mixing in stars. An example is Helio- and Asteroseismology looks at the pulsation of stars due to the formation of standing waves within the interior (Michel et al. 2006). By recording the fluctuations in spectrums of stars, the internal structure of a star can be inferred. Asteroseismology can be connected to convective boundary mixing and as a result a bestfit overshooting length can be estimated (Deheuvels, S. et al. 2010). However the amplitude of these oscillations are tiny, a few ppm for photometry and tens of cm/s for radial velocity, meaning measurements have a high associated uncertainty (Michel et al. 2006). Along with overshooting, it is possible to analyse the internal

rotational structure of the sun (Couvidat et al. 2003).

Another observation is massive eclipsing binaries. By measuring the Doppler shift in the spectral lines, the masses of the stars can be obtained. From the dimming in the light curve as one star eclipses the other, the radii can be obtained. Finally from the spectra the effective temperature can be read. From these fundamental parameters, stellar models can be fitted to a high precision, and it has been shown that including overshooting in models leads to better fits to observations (Baraffe et al. 2023).

The Hertzsprung-Russell diagram (HRD) is a common way of plotting a stars state, and as such characterising the stars into evolutionary groups. A common way of plotting it is to compare Luminosity of a star versus effective temperature, T_{eff} . A spectroscopic HRD (sHRD) instead replaces the stellar luminosity with $\mathcal{L} = T_{\text{eff}}^4/g$, where g is the surface gravity of the star. The power of this is that the sHRD has no reliance on distance and extinction of received light. This is because effective temperature can be determined from the spectral lines and surface gravity leads to broadening in the lines.

3 Experimental method

3.1 Extreme Value Theory

Convective boundary mixing is dominated by the rarest, powerful penetrative plumes. Figure 2 demonstrates that because the extreme events determine the deepest mixing, simple means of penetration depth fails to capture the full extent of overshooting (Pratt et al. 2017).

Extreme value theory (EVT) is an area of statistics, which attempts to describe the distribution of rare events (Gomes & Guillou 2015). It does this by considering the probability of maximum events over time (Pratt et al. 2017).

An important result is the generalised extreme value distribution (GEVD), which has a cumulative distribution function of

$$F(x) = \exp\left(-\left(1 + \kappa\left(\frac{x - \mu}{\lambda}\right)\right)^{-1/\kappa}\right), \quad (4)$$

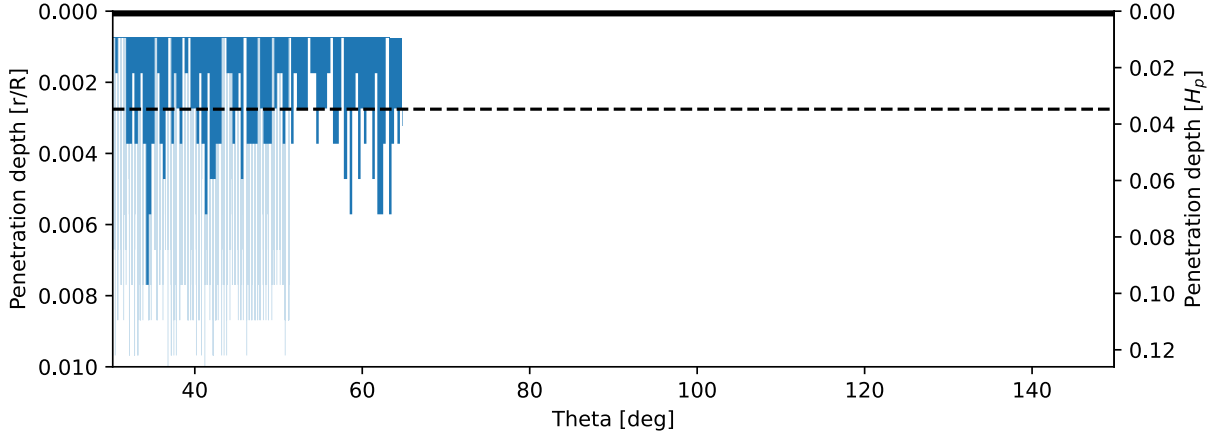


Figure 2: Angular structure of the penetration layer at an arbitrary time in 3D MUSIC simulation of the current Sun. The penetration depth in this illustration is determined by zeros of the vertical kinetic energy flux. The Schwarzschild criterion is indicated by the solid black line. A dashed black line indicates the average penetration depth at this time.

where μ is the location parameter, and λ is the scale parameter. The shape parameter or extreme value index (EVI) κ determines the functional extreme value distribution, with Weibull, $\kappa < 0$, Gumbel, $\kappa = 0$, and Fréchet, $\kappa > 0$. Pratt et al. (2017) found that plumes can be categorised into weak and strong events. Then via equation (4), Pratt observed for high resolution simulations the strong plumes can be modelled by a Gumbel distribution. The overshooting length can then be estimated as the location parameter, μ . A simplification in the method is to use the mean of maximal penetration depths to estimate overshooting (Baraffe, I. et al. 2021).

The use of EVT for CBM assumes no significant restructuring, that plumes are independent, and sufficient time-steps are simulated. The theory has been to define a diffusion coefficient (Pratt et al. 2017), discussed in Section 3.3, which was shown to over-predict mixing (Baraffe et al. 2017). However, EVT has also underpredicted overshooting when compared to sHRD and massive binary observations (Baraffe et al. 2023). The project will therefore analyse the validity in use of EVT around stellar boundaries.

3.2 Stellar simulations

Numerical simulations allow for the computation and analysis of the interior of stars, without direct observation and when no analytical solutions exist. Historically the first models were one-dimensional (1D) assuming spherical symmetry of the star. The low-dimensionality leads to lower computational requirements, meaning 1D models can be ran for much longer times than higher-dimensional alternatives. Therefore 1D models are primarily used for evolutionary models, capable of running from birth to death of stars.

However the simplification down to just radial dependence, means that parameters must be prescribed to these modelled based on inherent high-dimensional effects. Two-dimensional (2D) models, allow for the inclusion of 2D convection rolls, via including latitude into the model. The highest and most computationally intensive models are three-dimensional (3D), which include longitude, allowing for rotational effects to be studied.

The project will analyse 2D and 3D simulations of a $1M_{\odot}$ star, generated with the Multi-dimensional Stellar Implicit Code (MUSIC). A full description can be found in Viallet et al. (2011, 2016); Goffrey et al. (2017). MUSIC solves the inviscid Euler equations in presence

of external gravity and thermal diffusion:

$$\frac{\partial \rho}{\partial t} = -\vec{\nabla} \cdot (\rho \vec{v}) \quad (5)$$

$$\frac{\partial \rho \vec{v}}{\partial t} = -\vec{\nabla} \cdot (\rho \vec{v} \otimes \vec{v}) - \vec{\nabla} p + \rho \vec{g} \quad (6)$$

$$\frac{\partial \rho e}{\partial t} = -\vec{\nabla} \cdot (\rho e \vec{v}) - p \vec{\nabla} \cdot \vec{v} + \vec{\nabla} \cdot (\chi \vec{\nabla} T) + Q_{\text{nuc}} \quad (7)$$

where ρ is the density, $\vec{v}$ is the velocity, p is the gas pressure, $\vec{g}$ is the gravitational acceleration, e is the specific internal energy, χ is the thermal conductivity, T is the temperature and Q_{nuc} represents the nuclear energy rate. The symbol $\otimes$ is the outer product. In MUSIC, the gravitational acceleration is simplified to being radially symmetric, $\vec{g} = g(r)\hat{r}$, with the magnitude being recalculated each time-step using

$$g(r) = \frac{4\pi G}{r^2} \int_0^r \bar{\rho}(u) u^2 du \quad (8)$$

where $\bar{\rho}$ is the mean radial density profile.

MUSIC is a fully implicit model, which means it calculates all the variables simultaneously including their mutual dependence and variability at the future time step. The use of a fully implicit scheme allows for a larger time-step to be used, without loss of accuracy.

The timestep is variable, determined by advective, radiative, and hydrodynamic constraints via the Courant-Friedrichs-Lewy (CFL) condition (Viallet et al. 2011; Le Saux et al. 2025).

Simulations are initialized from a pre-calculated 1D MESA model (Paxton et al. 2010), which also defines the boundary conditions. Small perturbations are added to the initial state to accelerate relaxation. See Figure 3 for the process of using MUSIC within the project.

There are different ways to calculate the extent of penetration in the stellar simulations. Following an established method (for example Pratt et al. (2017); Baraffe et al. (2017)), the project will use two definitions. For other definitions see Dethero et al. (2024); Brun et al. (2017).

The first is where the vertical kinetic energy flux first equals zero past the convective boundary. The flux is defined by

$$F_{\text{kin}} = \frac{1}{2} \vec{v}_r \rho \vec{v}^2, \quad (9)$$

where ρ is the density and $\vec{v}$ is the velocity. The second is the first zero of vertical heat flux given by

$$F_c = \rho c_p \vec{v}_r \delta T \quad (10)$$

where c_p is the specific heat capacity at constant pressure, and δT is the temperature perturbation from the horizontal average of temperature.

3.3 321D link

While 3D simulations, such as MUSIC, provide detailed insights into the interior dynamics and processes, they are computational expensive and cannot be ran over evolutionary timescales. This limitation motivates the 321D link, the idea of using higher dimension simulations to inform and improve the lower, less computational intensive models (Pratt et al. 2020). To achieve this, 1D models incorporate free parameters which describe the higher dimensional physics' effect on the radial distributions.

In the case of convective boundary mixing a common procedure is to include an overshooting length in which increased mixing occurs (Pratt et al. 2017). This is done via a 1D diffusion coefficient, D , which in most simulations is used to describes multiple processes from molecular diffusion to convective motions (van Ballegooijen 1982). The Péclet number, defined as advective over diffusive transfer, can characterise which of these processes dominate the overall mixing (for example Pratt et al. (2017)). Earlier models used the form derived in Freytag et al. (1996), which used a low Pe regime as to match simulation data, given by

$$D_{\text{F96}} = v_B^2 \tau \exp(-2(r_B - r)/H_p) \quad (11)$$

where v_B is the rms radial velocity at the convective boundary, r_B , and the characteristic

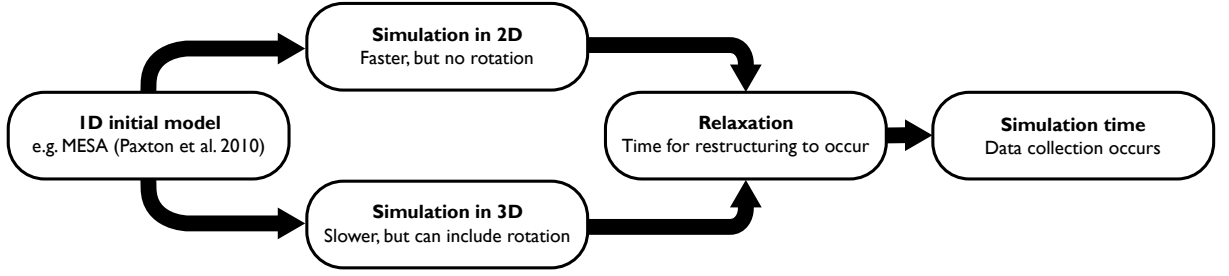


Figure 3: Schematic diagram of the process involved in running and utilising MUSIC.

timescale is given by $\tau = H_p/v_B$. However Freytag assumes a shallow convective zone and therefore according to Pratt et al. (2017) the correct regime is large Pe, and using a Gumbel distribution from EVT to fit simulation data. The calculation of diffusion coefficient can therefore be calculated from statistical analysis of 2D and 3D simulations to inform the extent of overshooting within a 1D model. To further this within the project, 2D and 3D MUSIC simulations of a $1M_\odot$ will be used (found in Vlaykov et al. (2022)). The models are derived from an initial 1D MESA model (for a description of MESA see Paxton et al. (2010)). The simulations can then be used to calculate the Péclet number and compare Pratt and Freytag to both 3D and 2D simulations of a current Sun. Potentially the value of the diffusion coefficient could also be used within new 1D MESA simulations and compared to observations.

Alongside this another link between 2D and 3D simulations can be examined, namely the numerical effect that 2D simulations give artificially higher velocities than their 3D counterparts (Pratt et al. 2020; Dethero et al. 2024). To do this the average velocity components will be compared, as seen in Figure 4 from Pratt et al. (2020), the values of non-dimensional numbers such as Prandtl, Peclet, Rayleigh, and Reynolds numbers between 2D, 3D, and expected Solar values.

3.4 Rotational effects

The rotation of stars evolves over its lifetime, allowing gyrochronology (Barnes 2003), via spectroscopy (Hartman et al. 2009). Conser-

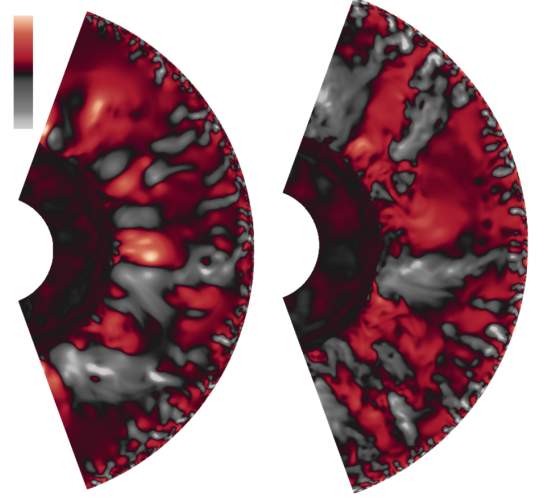


Figure 4: From Pratt et al. (2020) showing a snapshot of a radial velocity in a 2D simulation (left) and a slice in a 3D simulation (right), where red is flows outwards and grey inwards.

vation of angular momentum during collapse of molecular clouds, leads to solar-like stars starting on the main sequence (zero-age), have periods ranging from 0.2–8 days (Gallet, F. & Bouvier, J. 2015). Magnetic stellar winds slow rotation (Mestel 1984; Kawaler 1988). This magnetic activity comes from the interaction between rotation and convection (Brun & Browning 2017; Käpylä et al. 2023).

Rotation introduces the Coriolis force (Valis 2017), which diverts plumes perpendicular to the rotational axis, suppressing penetration depths (Brummell et al. 2002). Based on observed Lithium abundances a critical rotation rate of $\Omega_{\text{crit}} = 5\Omega_\odot$ has been proposed, above which mixing is significantly decreased Baraffe et al. (2017). However, this prescription was derived from 1D evolutionary models

rather than directly from hydrodynamic simulations, as such part of the project is to directly quantify the effect rotation has on overshooting from 3D simulations.

The non-dimensional Rossby number is a natural way to quantify the rotational dependence of overshooting (Brummell et al. 2002; Korre & Featherstone 2021). Due to observations giving a global rotation period of stars P_{rot} , it is common to define a global Rossby number,

$$\text{Ro} = \frac{P_{\text{rot}}}{\tau_{\text{conv}}} \quad (12)$$

where τ_{conv} is the convective turnover time. However, estimating τ globally is challenging (See et al. 2025). A local description can instead be calculated directly from simulations however, it also depends on the characteristic scale chosen. As such in the literature there are multiple differing definitions of the Rossby number (for example Korre & Featherstone (2021); See et al. (2025); Brummell et al. (2002); Breton et al. (2022)) In the project aims this uncertainty leads to comparing global and local Rossby numbers in describing overshooting lengths.

The Coriolis force also varies with latitude, with Taylor columns aligning flows perpendicular to rotation axis towards the poles, this has suggested overshooting also depends on latitude (Brummell et al. 2002; Brun et al. 2017). Another interpretation, is that convection is more vigorous at the equator and that causes the convective boundary to become prolate, whereas the overshooting remains spherical (Browning et al. 2004; Breton et al. 2022). This means the exact nature of the dependence on latitude is an open question.

The project will use simulations of three different rotational cases. A stationary case, $\Omega = 0$, which will be used to give a baseline for the overshooting length and compare to non-rotational models findings. A solar rotational case, $\Omega = \Omega_{\odot}$, will make it possible to compare to solar observations. Finally a fast rotating star, $\Omega = 5\Omega_{\odot}$, which is the same as the critical value found in Baraffe et al. (2017), and as such should show a suppressed pene-

tration length. Overall the dependence of the penetration length on rotation rate, can then be found for a solar-like star.

4 Conclusions

Taken together, this report illustrates how the mixing between convective and radiative zones significantly impacts the stellar observations and evolution, and the development of a more accurate description is needed to develop the 321D link.

From observations of the Sun and other stars, including massive binary systems, neither the Schwarzschild or Ledoux stability criteria are able to fully describe the derived internal structure of stars, and that the prescription of an overshooting length is required to bring simulations closer. The foundational text for overshooting in a stellar context is Zahn (1991), who found that penetration is dependent on the speed of motions within the fluid, see equation (3), however it does not account for the effect of rotation and the correct definition of the filling factor is disputed.

The project will apply the EVT, as described by Pratt et al. (2017), to 2D and 3D simulations to quantify how dimensionality and rotation affect convective boundary mixing. Attempting to identify a trend or critical rotation rate, as predicted by Baraffe et al. (2017). As well as comparing to observational data.

The results of this project are expected to therefore be to provide numerically motivated parametrisation of CBM for use in 1D stellar evolution code, such as MESA. By comparing the found regime to previous prescriptions, such as Freytag et al. (1996); Pratt et al. (2017), this project will contribute to the 321D link. Finally, the calibrated 1D models can be tested against observational data, therein connecting the numerical simulations to physical properties of stars, improving our understanding of their structure and evolution.

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